Chapter 24 Schrödinger Equation

24.1 Wave function

De Broglie's relation

$$E = \hbar\omega$$

$$\boldsymbol{p} = \hbar\boldsymbol{k}$$

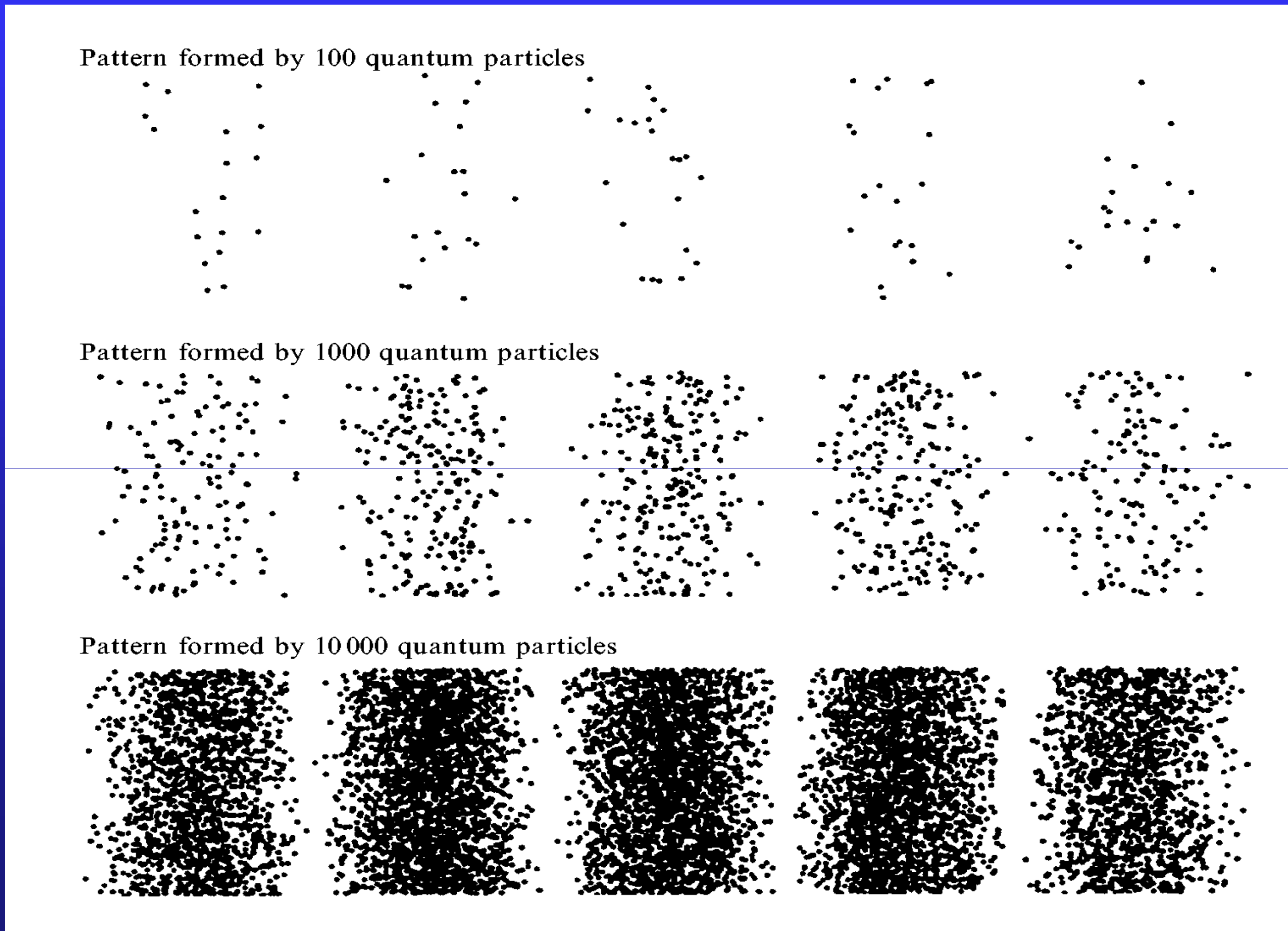
free particle $\longleftrightarrow$ plane wave

$$\psi(\boldsymbol{r},t) = Ae^{i(\boldsymbol{k}\cdot\boldsymbol{r}-\omega t)} = Ae^{\frac{i(\boldsymbol{p}\cdot\boldsymbol{r}-Et)}{\hbar}}$$

For an arbitrary particle $\psi(\boldsymbol{r},t)$ wave function

ψ Ψ psi the 23rd greek letter

What is the physical interpretation of the wave function?



$$\left|\Psi\right|^2 = \Psi^*\Psi \propto P(\boldsymbol{r})$$

Probability density

$$\left|\Psi\right|^2 d\tau = \Psi^*\Psi d\tau \propto P(\boldsymbol{r}) d\tau$$

Probability of a particle within the volume element $d\ \tau$

Max Born proposed a statistical interpretation about the wave function

- $|\Psi|^2 = \Psi^* \Psi$ is proportional to the probability density (* means complex conjugate).
- $|\Psi|^2 dx$ is proportional to the probability of finding the particle in an interval dx near x.
- $\int_{x_1}^{x_2} |\Psi(x)|^2 \mathrm{d}x$ is proportional to the probability of finding particle in the interval from x_1 to x_2.
- Ψ is probability amplitude.

Max Born shared the 1954 Nobel Prize for his fundamental research in quantum mechanics, especially for his statistical interpretation of the wave function.(Copenhagen interpretation)

According to Born's theory, wave function is assigned the following basic properties.

- Wave function is single-value, continuous, and finite.

- Normalization $1 = \int |\Psi|^2 \mathrm{d}\tau$. in most case is preferred though $|\Psi|^2 \mathrm{d}\tau$ and $C|\Psi|^2 \mathrm{d}\tau$ give out same relative probability.

- Principle of Superposition

if Ψ_1, Ψ_2 are possible states of the system, then $\Psi = C_1 \Psi_1 + C_2 \Psi_2$ is also a possible state of the system.

24.2 Schrödinger equation

From Felix Bloch’s Biographical Memoirs

Once at the end of a colloquium I heard Debye saying something like: "Schrödinger, you are not working right now on very important problems anyway. Why don't you tell us some time about that thesis of de Broglie, which seems to have attracted some attention?"

So in one of the next colloquia, Schrödinger gave a beautifully clear account of how de Broglie associated a wave with a particle and how he could obtain the quantization rules of Niels Bohr and Sommerfeld by demanding that an integer number of waves should be fitted along a stationary orbit.

When he had finished, Debye casually remarked that this way of talking was rather childish. As a student of Sommerfeld he had learned that, **to deal properly with waves, one had to have a wave equation.**

It sounded quite trivial and did not seem to make a great impression, but Schrödinger evidently thought a bit more about the idea afterwards.

Just a few weeks later he gave another talk in the colloquium which he started by saying: "**My colleague Debye suggested that one should have a wave equation; well I have found one!"** And then he told us essentially what he was about to publish under the title "Quantization as Eigenvalue Problem" as the first paper of a series in the Annalen der Physik(物理年鉴)．

The wave function of a free particle is written as

1D $$\Psi(x,t) = Ae^{i(k\cdot x - \omega t)} = Ae^{\frac{i}{\hbar}(p\cdot x - Et)}.$$

3D $$\Psi(\mathbf{r},t) = Ae^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)} = Ae^{\frac{i}{\hbar}(\mathbf{p}\cdot\mathbf{r} - Et)}.$$

$$\frac{\partial}{\partial t}\Psi = -\frac{i}{\hbar}E\Psi \qquad\qquad E \rightarrow i\hbar\frac{\partial}{\partial t}$$

$$\nabla\Psi = \frac{i}{\hbar}\mathbf{p}\Psi \qquad\qquad \mathbf{p} \rightarrow \frac{\hbar}{i}\nabla$$

$$\left(\frac{\hbar}{i}\nabla\right)\cdot\left(\frac{\hbar}{i}\nabla\right)\Psi = -\hbar^2\nabla^2\Psi = (\mathbf{p}\cdot\mathbf{p})\Psi = p^2\Psi$$

The relativistic energy-momentum relation is

$$E^2 = m_0^2 c^4 + p^2 c^2.$$

In the non-relativistic limit, we have

$$E = \sqrt{m_0^2 c^4 + p^2 c^2} = m_0 c^2 \sqrt{1 + \frac{p^2}{m_0^2 c^2}}$$

$$\approx m_0 c^2 \left(1 + \frac{1}{2}\frac{p^2}{m_0^2 c^2}\right) = m_0 c^2 + \frac{p^2}{2m}$$

Thus, the non-relativistic energy-momentum relation is

$$E = \frac{p^2}{2m}$$

$$\frac{\partial}{\partial t}\Psi = -\frac{i}{\hbar}E\Psi$$

$$i\hbar\frac{\partial}{\partial t}\Psi = E\Psi$$

$$-\hbar^2\nabla^2\Psi = p^2\Psi$$

$$-\frac{\hbar^2}{2m}\nabla^2\Psi = \frac{p^2}{2m}\Psi$$

$$E = \frac{p^2}{2m}$$

$$i\hbar\frac{\partial}{\partial t}\Psi = -\frac{\hbar^2}{2m}\nabla^2\Psi$$

Wave equation for free particle

In a potential field

$$E = \frac{p^2}{2m} + U(\boldsymbol{r})$$

$$i\hbar\frac{\partial}{\partial t}\Psi = -\frac{\hbar^2}{2m}\nabla^2\Psi + U(\boldsymbol{r})\Psi$$

Schrödinger equation

*For relativistic case, we have Klein-Gorden equation and Dirac equation

From the change rate of probability density, we can find the probability current density.

$$\frac{\partial}{\partial t}\left[\Psi^*(\boldsymbol{r},t)\Psi(\boldsymbol{r},t)\right]=\frac{\partial\Psi^*}{\partial t}\Psi+\Psi^*\frac{\partial\Psi}{\partial t}$$

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\Psi=-\frac{\hbar^2}{2m}\nabla^2\Psi+U\Psi$$

$$-i\hbar\frac{\partial}{\partial t}\Psi^*=-\frac{\hbar^2}{2m}\nabla^2\Psi^*+U\Psi^*$$

$$\frac{\partial}{\partial t}\left|\Psi\right|^{2}=-\frac{\hbar}{2im}\left[\Psi^{*}\nabla^{2}\Psi-\Psi\nabla^{2}\Psi^{*}\right]$$

$$=-\frac{\hbar}{2im}\nabla\cdot\left[\Psi^{*}\nabla\Psi-\Psi\nabla\Psi^{*}\right]$$

$$\equiv-\nabla\cdot j$$

$$\nabla\cdot(\varphi\, \boldsymbol{f})=\nabla\varphi\cdot\boldsymbol{f}+\varphi\nabla\cdot\boldsymbol{f}$$

$$\nabla\cdot\left(\Psi^{*}\nabla\Psi\right)=\nabla\Psi^{*}\cdot\nabla\Psi+\Psi^{*}\nabla^{2}\Psi$$

$$\nabla\cdot\left(\Psi\nabla\Psi^{*}\right)=\nabla\Psi\cdot\nabla\Psi^{*}+\Psi\nabla^{2}\Psi^{*}$$

The probability current density is

$$\boldsymbol{j} = \frac{\hbar}{2\mathrm{i}m}\left[\Psi^{*}\nabla\Psi - \Psi\nabla\Psi^{*}\right]$$

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\boldsymbol{j} = 0$$

the continuity condition

$$\frac{\partial}{\partial t}\left|\Psi\right|^{2} = -\frac{\hbar}{2\mathrm{i}m}\nabla\cdot\left[\Psi^{*}\nabla\Psi - \Psi\nabla\Psi^{*}\right]$$

To integrate over all space

$$\frac{\partial}{\partial t}\int_{\infty}\left|\Psi\right|^{2}d\tau = -\frac{\hbar}{2\mathrm{i}m}\oint_{\infty}\left[\Psi^{*}\nabla\Psi - \Psi\nabla\Psi^{*}\right]\cdot d\boldsymbol{S} = 0$$

If the potential is explicitly time-independent , it is possible to express the solution of the Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}\Psi=-\frac{\hbar^2}{2m}\nabla^2\Psi+U(\boldsymbol{r})\Psi$$

as the product: $\Psi(\boldsymbol{r},t)=\psi(\boldsymbol{r})f(t).$

Substitute the expression into the equation, we have

$$i\hbar\psi(\boldsymbol{r})\frac{df(t)}{dt}=-\frac{\hbar^2}{2m}\left[\nabla^2\psi(\boldsymbol{r})\right]f(t)+U(\boldsymbol{r})\psi(\boldsymbol{r})f(t)$$

or

$$\frac{i\hbar}{f(t)}\frac{df(t)}{dt}=\frac{1}{\psi(\boldsymbol{r})}\left[-\frac{\hbar^2}{2m}\nabla^2\psi(\boldsymbol{r})+U(\boldsymbol{r})\psi(\boldsymbol{r})\right]$$

$$\frac{i\hbar}{f(t)}\frac{df(t)}{dt} = \frac{1}{\psi(\boldsymbol{r})}\left[-\frac{\hbar^2}{2m}\nabla^2\psi(\boldsymbol{r}) + U(\boldsymbol{r})\psi(\boldsymbol{r})\right]$$

function of t

function of $\boldsymbol{r}$

If the equality is to hold for all values of $\boldsymbol{r}$ and for all values of t, both the L.H.S. and the R.H.S must be equal to the same constant E.

$$\frac{i\hbar}{f(t)}\frac{df(t)}{dt} = E$$

separation constant

$$\frac{1}{\psi(\boldsymbol{r})}\left[-\frac{\hbar^2}{2m}\nabla^2\psi(\boldsymbol{r}) + U(\boldsymbol{r})\psi(\boldsymbol{r})\right] = E$$

$$f(t) = Ce^{-\frac{i}{\hbar}Et}$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\boldsymbol{r}) + U(\boldsymbol{r})\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r})$$

stationary Schrödinger equation.

After defining Hamiltonian operator

$$\hat{H} \equiv -\frac{\hbar^2}{2m}\nabla^2 + U(\boldsymbol{r})$$

the stationary Schrödinger equation can be rewritten as

$$\hat{H}\psi = E\psi.$$

$$\hat{H}\psi = E\psi.$$

The equation is called eigenvalue equation of the operator $\hat{H}$. Allowed value(s) of constant E is (are) eigenvalue(s) and the solution is called eigenfunction(s).

If all eigenvalues E and eigenfunctions $\psi_E(\boldsymbol{r})$ are found, the general solution of the Schrödinger equation can be written as

$$\Psi(\boldsymbol{r},t) = \sum_E C(E)\psi_E(\boldsymbol{r})e^{-\frac{\mathrm{i}}{\hbar}E\,t}$$

The expectation value of position is

$$\langle x \rangle = \int \psi^* x \psi \, \mathrm{d}x \qquad \text{1D}$$

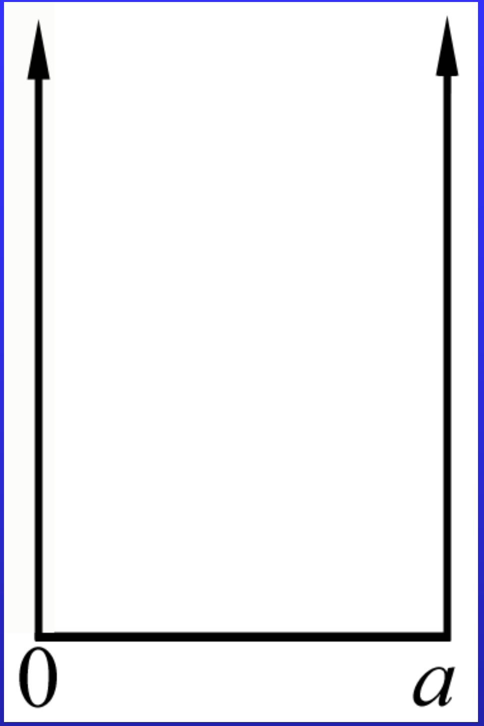
$$\langle \boldsymbol{r} \rangle = \int \psi^* \boldsymbol{r} \psi \, \mathrm{d}\tau \qquad \text{3D}$$

The expectation value of momentum is

$$\langle p \rangle = \int \psi^* \frac{\hbar}{i} \frac{\mathrm{d}}{\mathrm{d}x} \psi \, \mathrm{d}x \qquad \text{1D}$$

$$\langle \boldsymbol{p} \rangle = \int \psi^* \frac{\hbar}{i} \nabla \psi \, \mathrm{d}\tau \qquad \text{3D}$$

24.3 One dimensional potential well



$$U(x)=\begin{cases} 0\,, & 0<x<a \\ \infty, & x\le 0, x\ge a \end{cases}$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}=E\psi\,(0<x<a)$$

$$\psi=0 \ \ \text{elsewhere}$$

$$\frac{d^2\psi}{dx^2}+k^2\psi=0 \qquad \left(k^2\equiv\frac{2mE}{\hbar^2}\right)$$

The general solution is simply

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

With boundary conditions

$$\psi(0) = \psi(a) = 0$$

the solution is

$$\psi_n(x) = A\sin(k_n x), k_n = \frac{n\pi}{a}, \qquad (n = 1,2,\cdots)$$

the energy eigenvalues

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{a}\right)^2 \qquad k^2 \equiv \frac{2mE}{\hbar^2}$$

For a = 1nm, E_1 and spacing are

$$E_1 = \frac{\hbar^2 \pi^2}{2ma^2} = \frac{h^2 c^2}{8mc^2 a^2}$$

$$= \frac{(1240\,\mathrm{eV} \cdot 10^{-9}\,\mathrm{m})^2}{8 \times 0.511 \times 10^6\,\mathrm{eV} \times 10^{-18}\,\mathrm{m}^2}$$

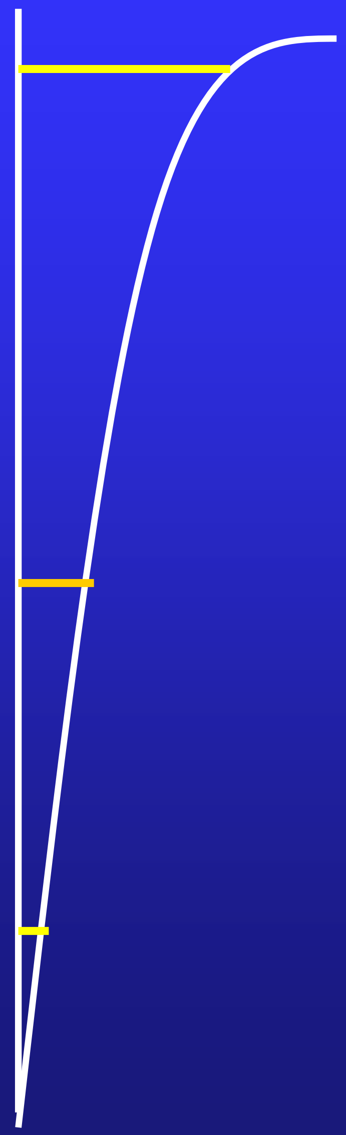
$$= 0.376\,\mathrm{eV}$$

$$\Delta E_n = E_{n+1} - E_n = E_1[(n+1)^2 - (n)^2]$$

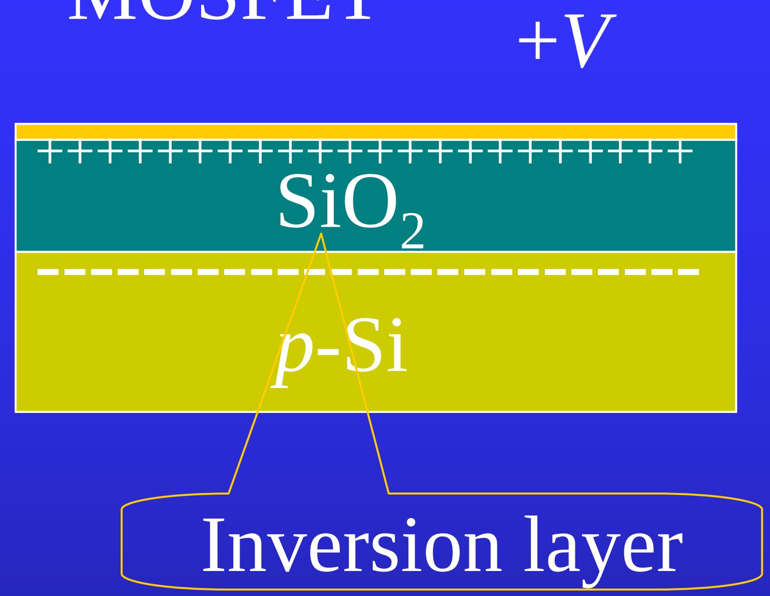
$$= (2n+1)E_1 \approx 2nE_1,$$

$$\Delta E_n = 2nE_1 \sim 2na^{-2} \qquad \frac{\Delta E_n}{E_n} = \frac{2}{n}$$

2D electron gas system



MOSFET



MOSFET ---metallic oxide semiconductor field effect transistor

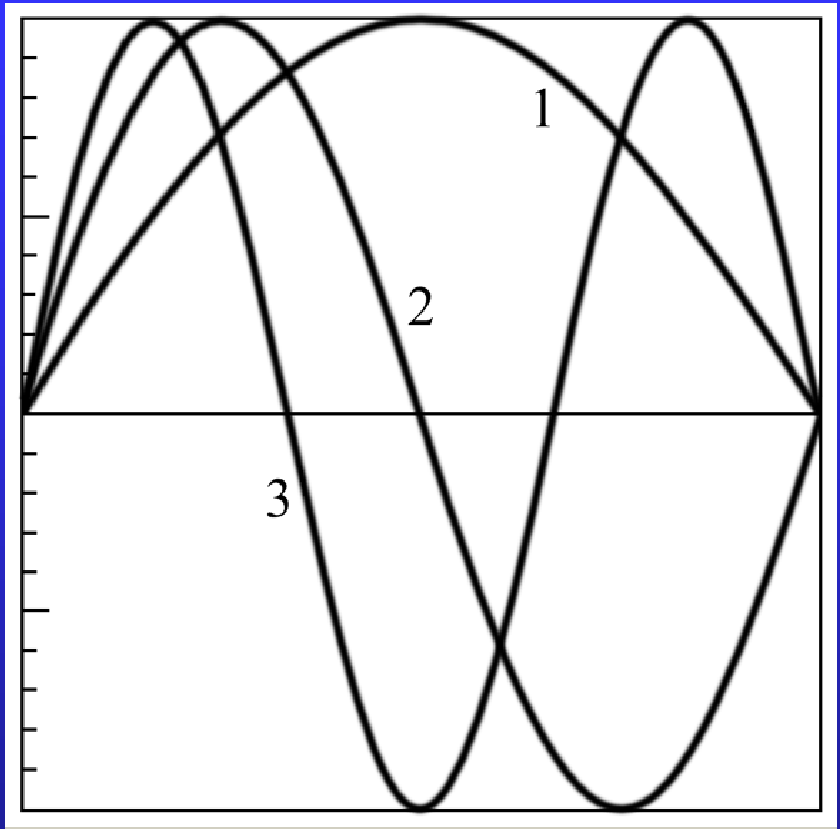
Constant A can be fixed through normalization condition, i.e.,

$$1 = A^2 \int \sin^2\left(\frac{n\pi x}{a}\right) dx.$$

The normalized wavefunctions are

$$\psi_n(x) = \begin{cases} \sqrt{\dfrac{2}{a}} \sin\left(\dfrac{n\pi x}{a}\right) & 0 < x < a \\ 0 & x \le 0, x \ge a \end{cases}.$$

$$E_n = \frac{\hbar^2 \pi^2}{2ma^2} n^2 \qquad (n = 1, 2, \cdots)$$



The probability density

$$|\psi_n(x)|^2 \mathrm{d}x$$

in classical case,

$$\frac{dt}{T} = \frac{\dfrac{2dx}{v}}{\dfrac{2a}{v}} = \frac{2dx}{2a} = \frac{dx}{a}.$$

The stationary wave functions are orthonormalized

$$\int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi x}{a}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) dx = \delta_{mn} = \begin{cases} 0 \ , & m \neq n \\ 1 \ , & m = n \end{cases}$$

Including time-dependent factor, the wave functions are

$$\Psi_n(x,t)=\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi a}{a}\right)e^{-\frac{i}{\hbar}E_n t}$$

$$=\frac{1}{\sqrt{2a}}\left[e^{\frac{i}{\hbar}\left(\frac{n\pi x}{a}-E_n t\right)}-e^{-\frac{i}{\hbar}\left(\frac{n\pi x}{a}+E_n t\right)}\right]$$

Example Show that the expectation value of x is $a/2$, independent of the quantum state.

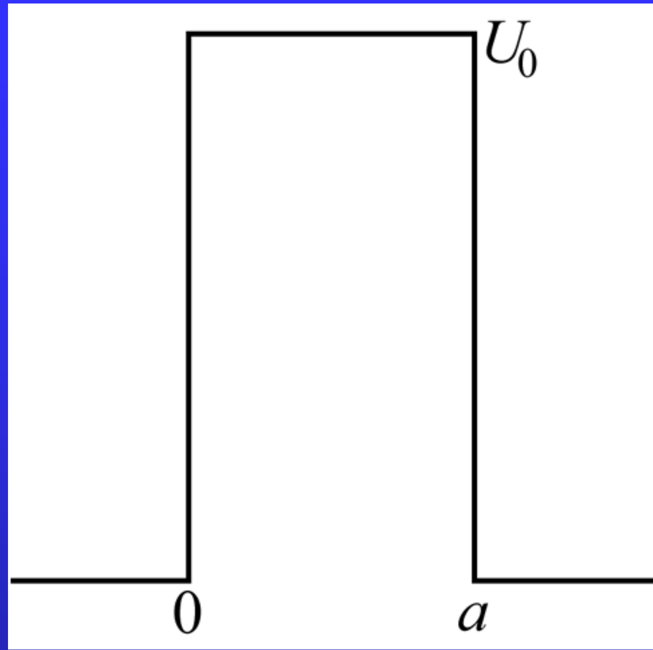
Solution:

$$
\begin{aligned}
\langle x \rangle &= \int_0^a \psi_n^*(x)\, x\, \psi_n(x)\, dx \\
&= \frac{2}{a}\int_0^a x \sin^2 \frac{n\pi x}{a} dx \\
&= \frac{2}{a}\int_0^a x \frac{1}{2}\left(1-\cos\frac{2n\pi x}{a}\right) dx \\
&= \frac{2}{a}\left[\frac{x^2}{4}\right]_0^a = \frac{a}{2}.
\end{aligned}
$$

Assignment: 24.3, 24.4

24.4 Potential barrier

$$U(x)=\begin{cases} U_0, & 0<x<a \\ 0, & x\le 0, x\ge a \end{cases}$$



In classical case, only particle of energy higher than U_0 can go over the barrier.

The stationary Schrödinger equations are

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi = E\psi \qquad (x<0, x>a)$$

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi + U_0\psi = E \quad (0<x<a)$$

Define (for $U_0>E$)

$$k^2 = \frac{2mE}{\hbar^2}, \quad \kappa^2 = \frac{2m(U_0 - E)}{\hbar^2}.$$

We have

$$\psi'' + k^2\psi = 0 \ \ (x<0, x>a),$$

$$\psi'' - \kappa^2\psi = 0 \ \ (0<x<a).$$

The solution is

reflected

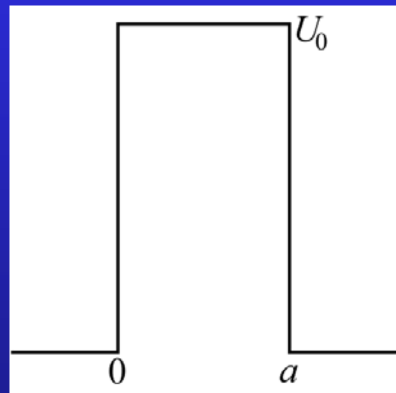
incident

$$\psi_1 = Ae^{i\,kx} + Be^{-i\,kx} \quad (x<0)$$

$$\psi_2 = Ce^{\kappa x} + De^{-\kappa x} \quad (0<x<a)$$

$$\psi_3 = Ee^{i\,kx} + \cancel{Fe^{-i\,kx}} \quad (x>a).$$

transmitted



$$f(t) = Ce^{-\frac{i}{\hbar}E\,t}$$

$$ikx - \frac{i}{\hbar}Et = i\left(kx - \frac{1}{\hbar}Et\right)$$ propagating to the right

The boundary conditions are

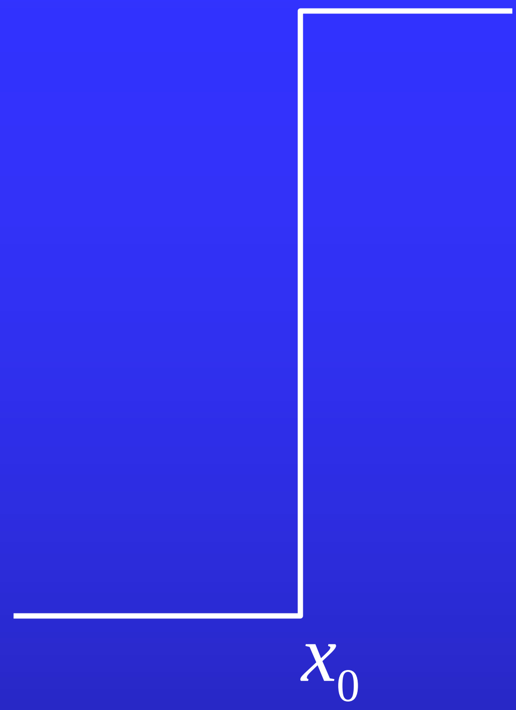
$$\psi_1\big|_{x=0} = \psi_2\big|_{x=0} \quad \rightarrow A + B = C + D,$$

$$\left.\frac{d\psi_1}{dx}\right|_{x=0} = \left.\frac{d\psi_2}{dx}\right|_{x=0} \quad \rightarrow ik(A - B) = \kappa(C - D),$$

$$\psi_2\big|_{x=a} = \psi_3\big|_{x=a} \quad \rightarrow Ce^{\kappa a} + De^{-\kappa a} = Ee^{ika},$$

$$\left.\frac{d\psi_2}{dx}\right|_{x=a} = \left.\frac{d\psi_3}{dx}\right|_{x=a} \quad \rightarrow \kappa(Ce^{\kappa a} - De^{-\kappa a}) = ikEe^{ika}.$$

$$\frac{d^2}{dx^2}\psi(x) = \frac{2m}{\hbar^2}\left(U(x) - E\right)\psi(x)$$



$$\int_{x_0-\varepsilon}^{x_0+\varepsilon} \frac{d^2}{dx^2}\psi(x)dx = \int_{x_0-\varepsilon}^{x_0+\varepsilon} \frac{2m}{\hbar^2}\left(U(x) - E\right)\psi(x)dx$$

$$\left.\frac{d\psi(x)}{dx}\right|_{x_0+\varepsilon} - \left.\frac{d\psi(x)}{dx}\right|_{x_0-\varepsilon} = \int_{x_0-\varepsilon}^{x_0+\varepsilon} \frac{2m}{\hbar^2}\left(U(x) - E\right)\psi(x)dx$$

From these equations, we obtain

$$|A|^2 = \frac{|E|^2[(\kappa^2 + k^2)^2(e^{-\kappa a} - e^{\kappa a})^2 + 16\kappa^2 k^2]}{16\kappa^2 k^2},$$

$$|B|^2 = \frac{|E|^2(\kappa^2 + k^2)^2(e^{-\kappa a} - e^{\kappa a})^2}{16\kappa^2 k^2}.$$

The reflection and transmission coefficients are

$$R = \frac{|B|^2}{|A|^2}, \quad T = \frac{|E|^2}{|A|^2}.$$

It can be calculated that

$$T = \frac{|E|^2}{|A|^2} = \frac{16\kappa^2 k^2}{(\kappa^2 + k^2)^2 (e^{-\kappa a} - e^{\kappa a})^2 + 16\kappa^2 k^2}$$

$$\underset{\kappa a >> 1}{\approx} \frac{16\kappa^2 k^2}{(\kappa^2 + k^2)^2} e^{-2\kappa a}.$$

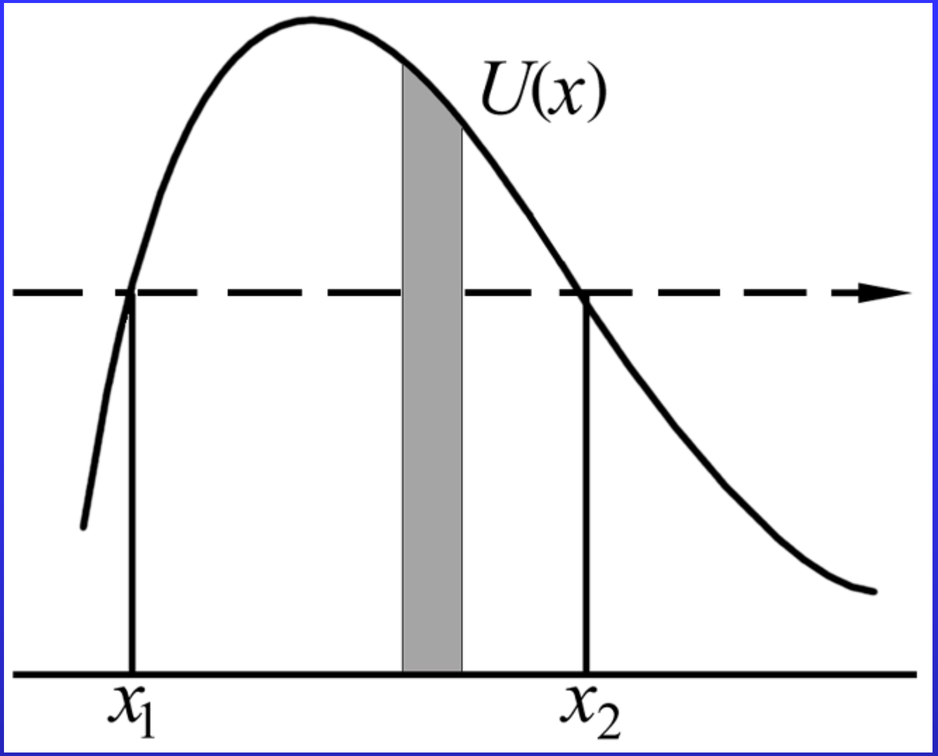
Even for $E < U_0$, we still can find finite T. This is called tunneling effect. Noting that

$$\kappa = \frac{\sqrt{2m(U_0 - E)}}{\hbar},$$

we have

$$T \approx T_0 \exp\left(-\frac{2}{\hbar}\sqrt{2m\left(U_0 - E\right)}a\right).$$

Example 24.1

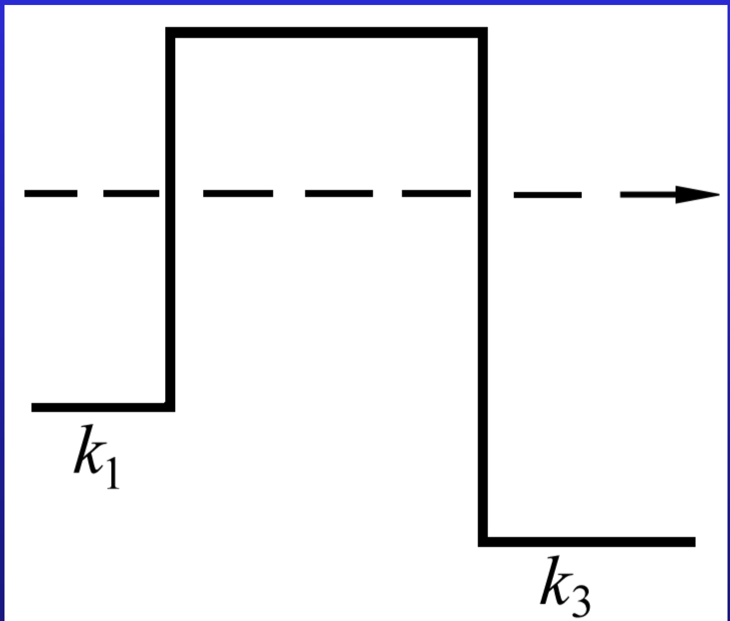


$$T = T_0 \exp\left(-\frac{2}{\hbar} \int_{x_1}^{x_2} \sqrt{2m\left(U(x) - E\right)}\,\mathrm{d}x \right)$$

Exact definition:

$$R = \frac{|j_R|}{|j_I|}, \qquad T = \frac{|j_T|}{|j_I|}.$$

For example tunnel diode (also called Esaki diode)



$$\psi_I = A e^{i k_1 x} \qquad k_1^2 = \frac{2m(E - U_1)}{\hbar^2}$$

$$\psi_R = B e^{-i k_1 x}$$

$$\psi_T = E e^{i k_3 x} \qquad k_3^2 = \frac{2m(E - U_3)}{\hbar^2}$$

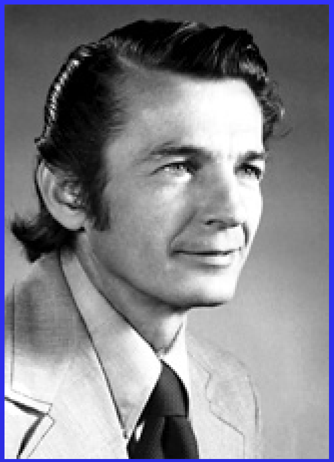
$$j_R = \frac{1}{2im}\left[\psi_R^* \frac{d\psi_R}{dx} - \frac{d\psi_R^*}{dx}\psi_R\right]$$

$$= \frac{i\hbar}{2m}\left[|B|^2(ik_1) - |B|^2(-ik_1)\right]$$

$$= -|B|^2\frac{\hbar k_1}{m}$$

$$j_I = |A|^2\frac{\hbar k_1}{m}, \qquad j_T = |E|^2\frac{\hbar k_3}{m} \ .$$

$$T = \frac{|E|^2 k_3}{|A|^2 k_1}$$



Leo Esaki(江崎)	Semiconductor
Ivar Giaever	Superconductor
Brian D. Josephson	Josephson effect

1973 Noble laureates

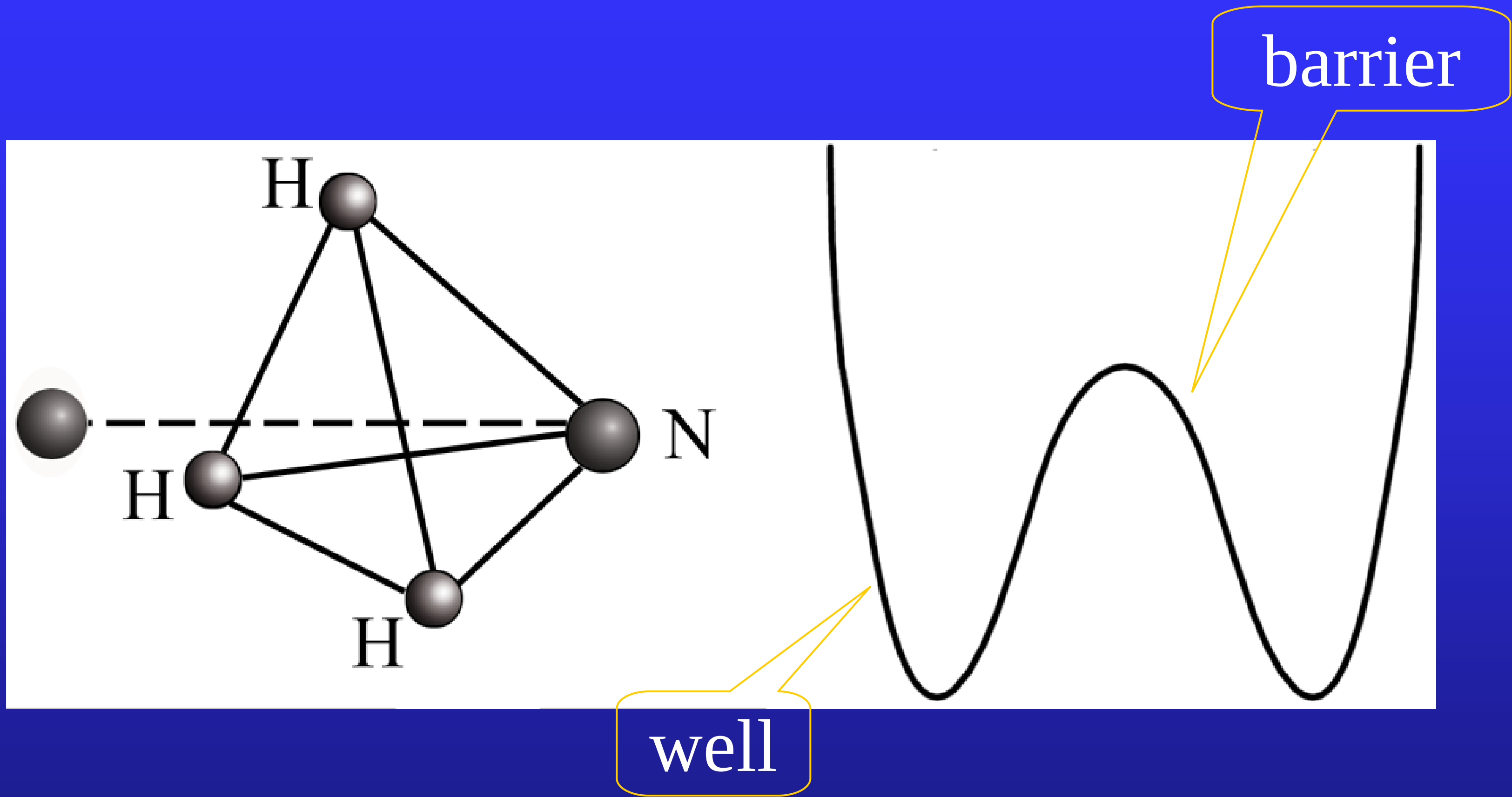
While first two found tunneling in semiconductor and superconductor respectively, the latter predicted theoretically properties of super-current through a tunnel barrier.

Brian David Josephson demonstrated that in a situation when there is electron flow between two superconductors through an insulating layer (in the absence of an applied voltage), and a voltage is applied, the current stops flowing and oscillates at a high frequency.

The Josephson effect is influenced by magnetic fields in the vicinity, a capacity that enables the Josephson junction to be used in devices that measure extremely weak magnetic fields, such as superconducting quantum interference devices (SQUIDs).

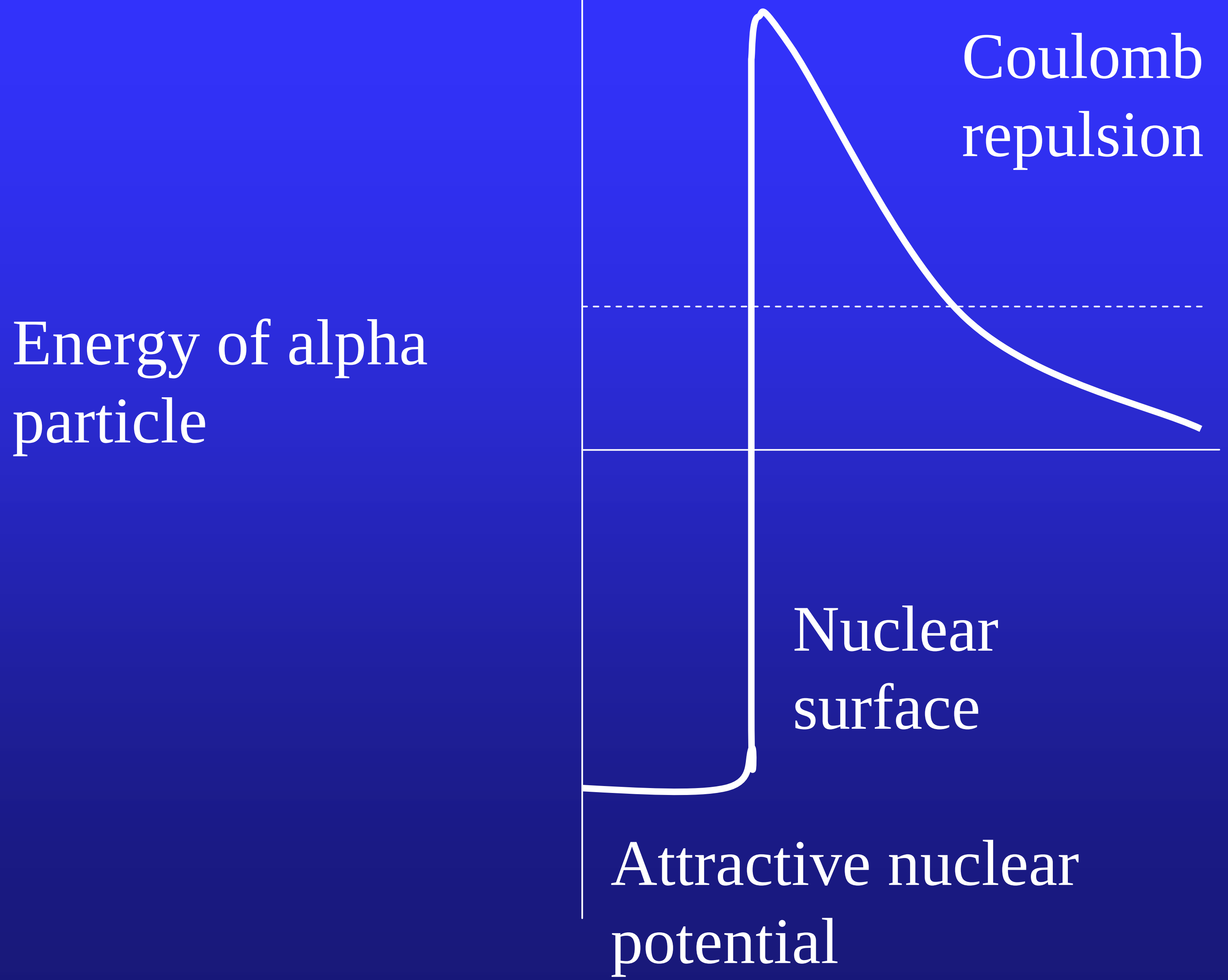
Josephson junction, Josephson tunneling, Josephson current

Ammonia inversion



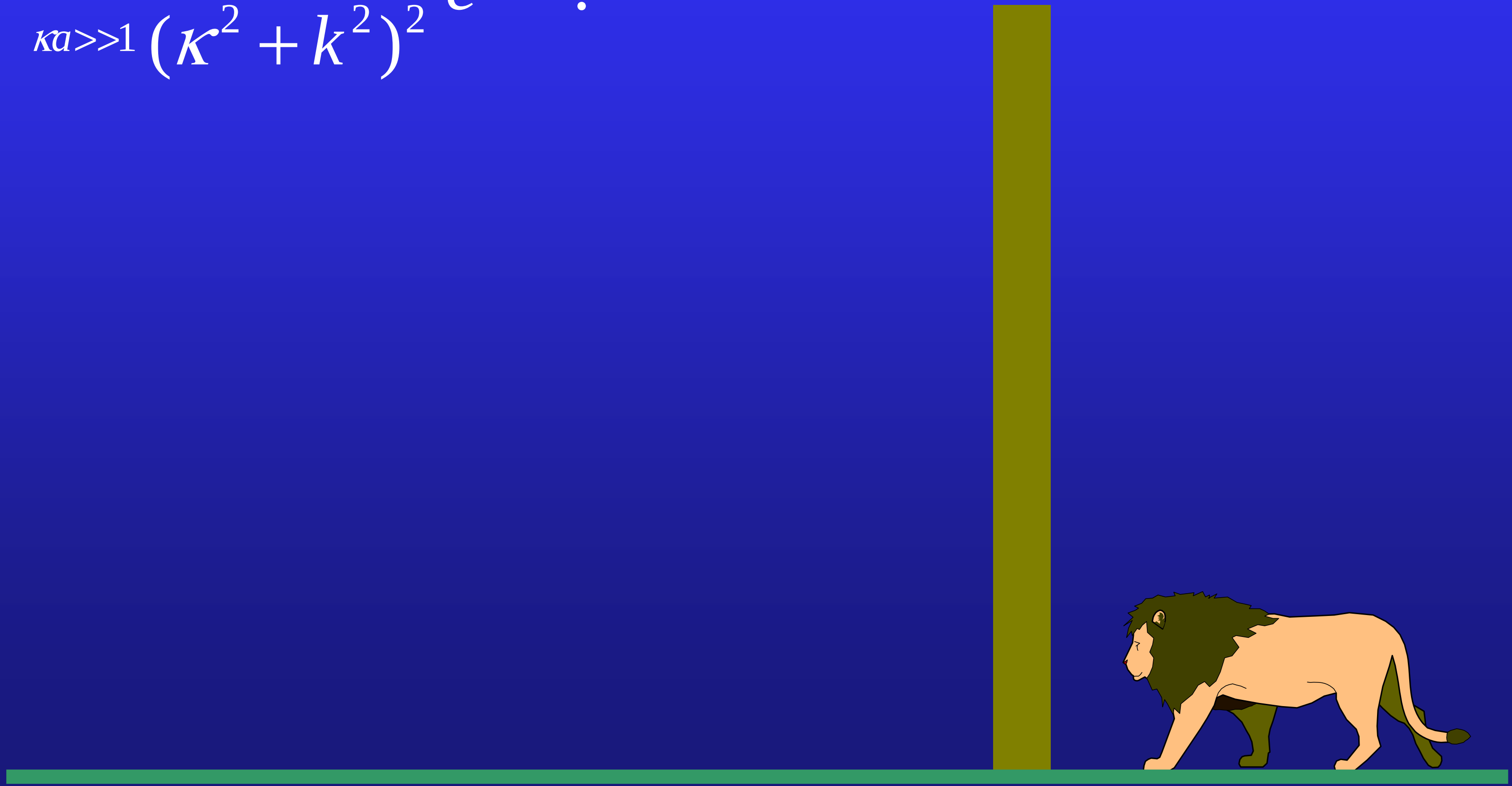
10^{10} Hz Quantum amplifier

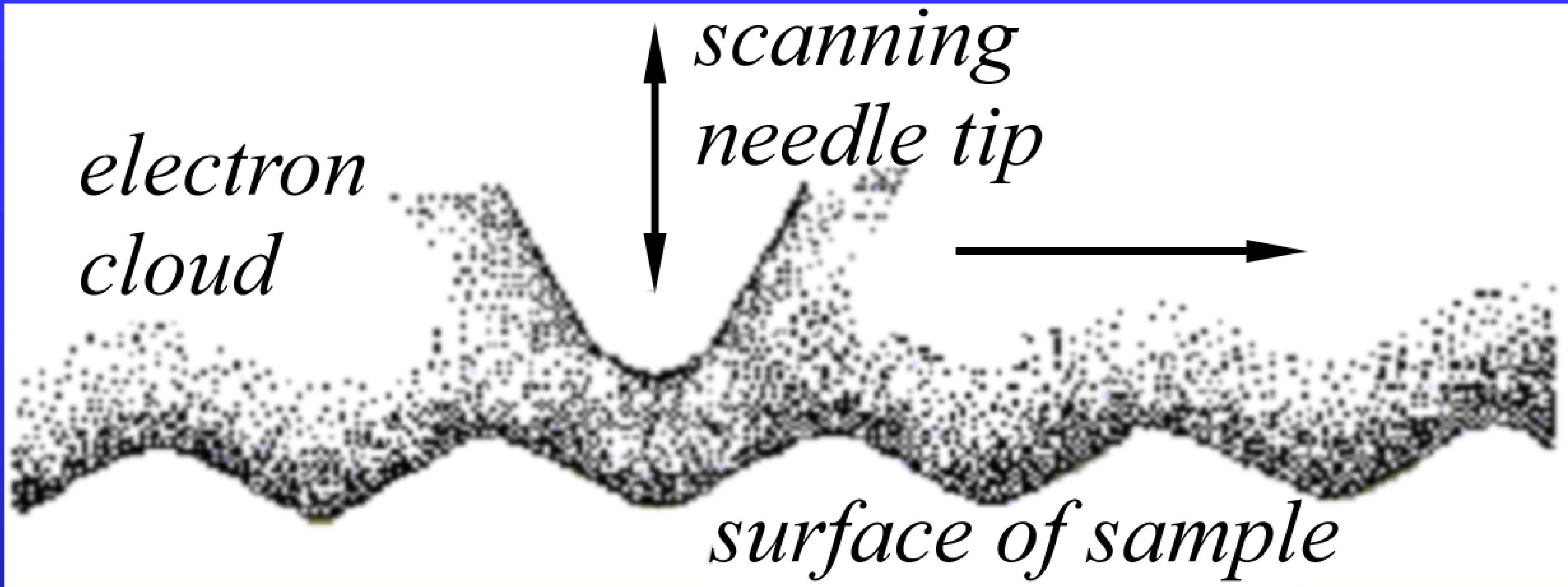
Alpha decay

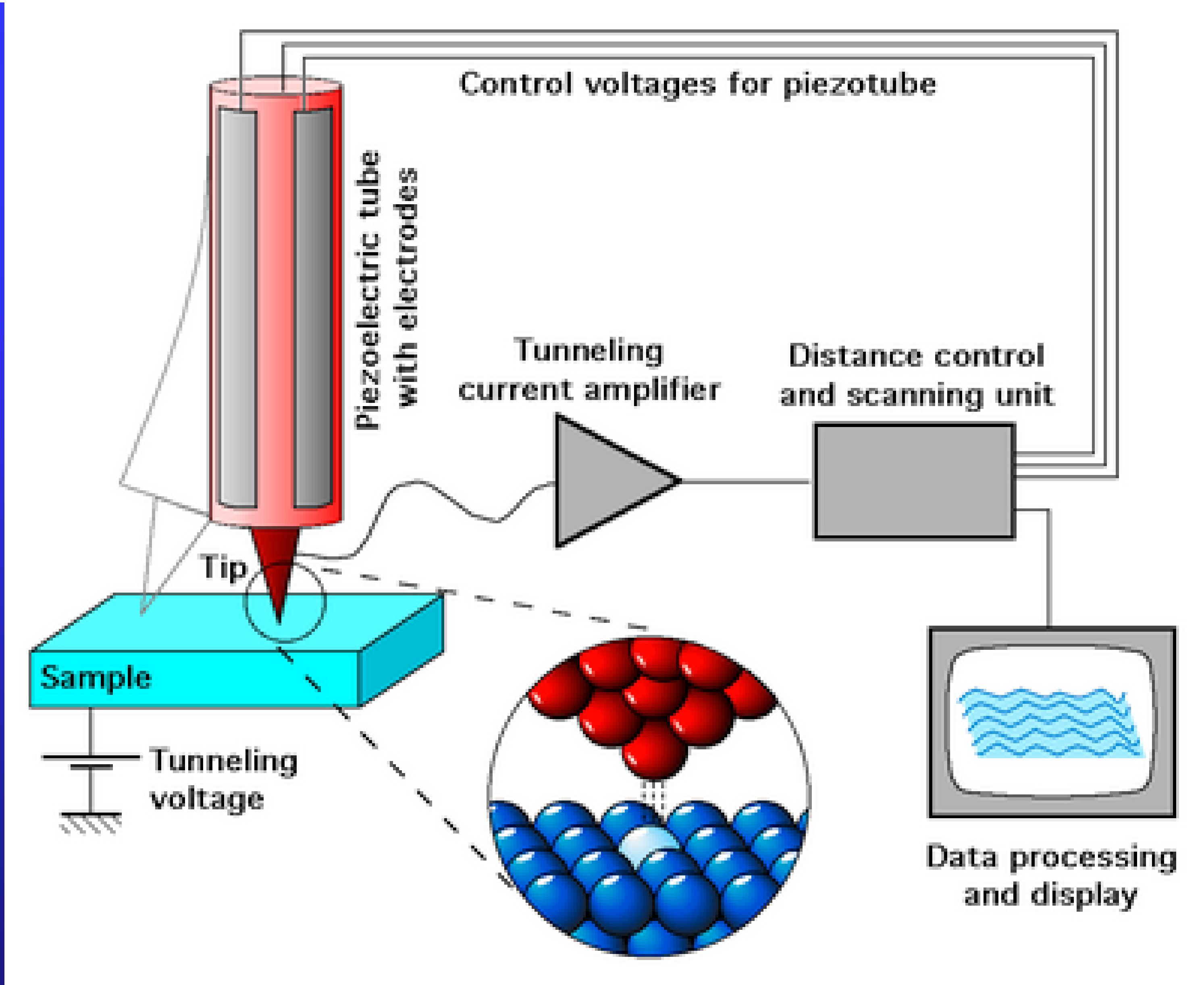

Coulomb repulsion
Energy of alpha particle
Nuclear surface
Attractive nuclear potential

$$T = \frac{|E|^2}{|A|^2} = \frac{16\kappa^2 k^2}{(\kappa^2 + k^2)^2 (e^{-\kappa a} - e^{\kappa a})^2 + 16\kappa^2 k^2}$$

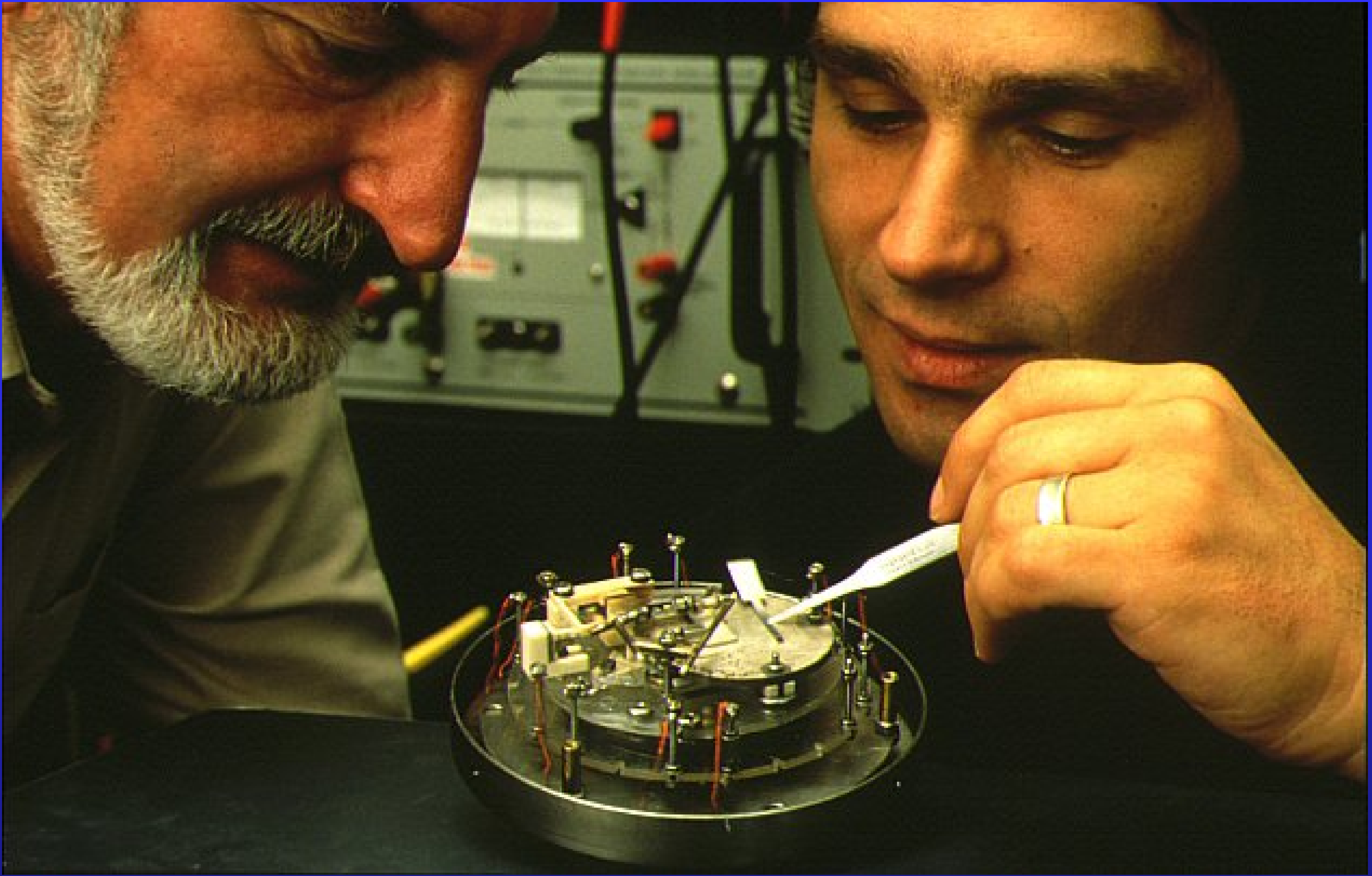
$$\underset{\kappa a >> 1}{\approx} \frac{16\kappa^2 k^2}{(\kappa^2 + k^2)^2} e^{-2\kappa a}.$$



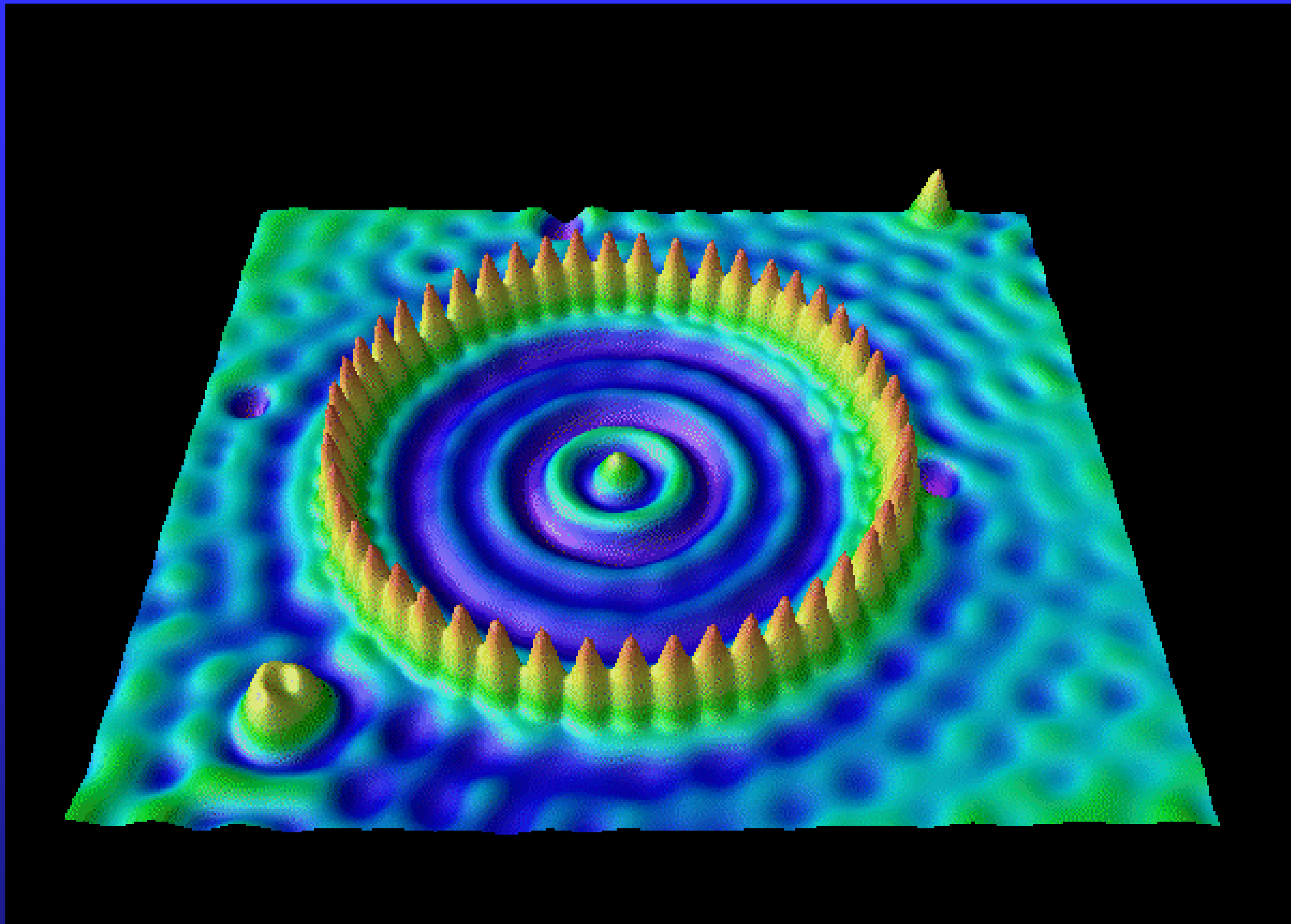

scanning
needle tip
electron
cloud
surface of sample



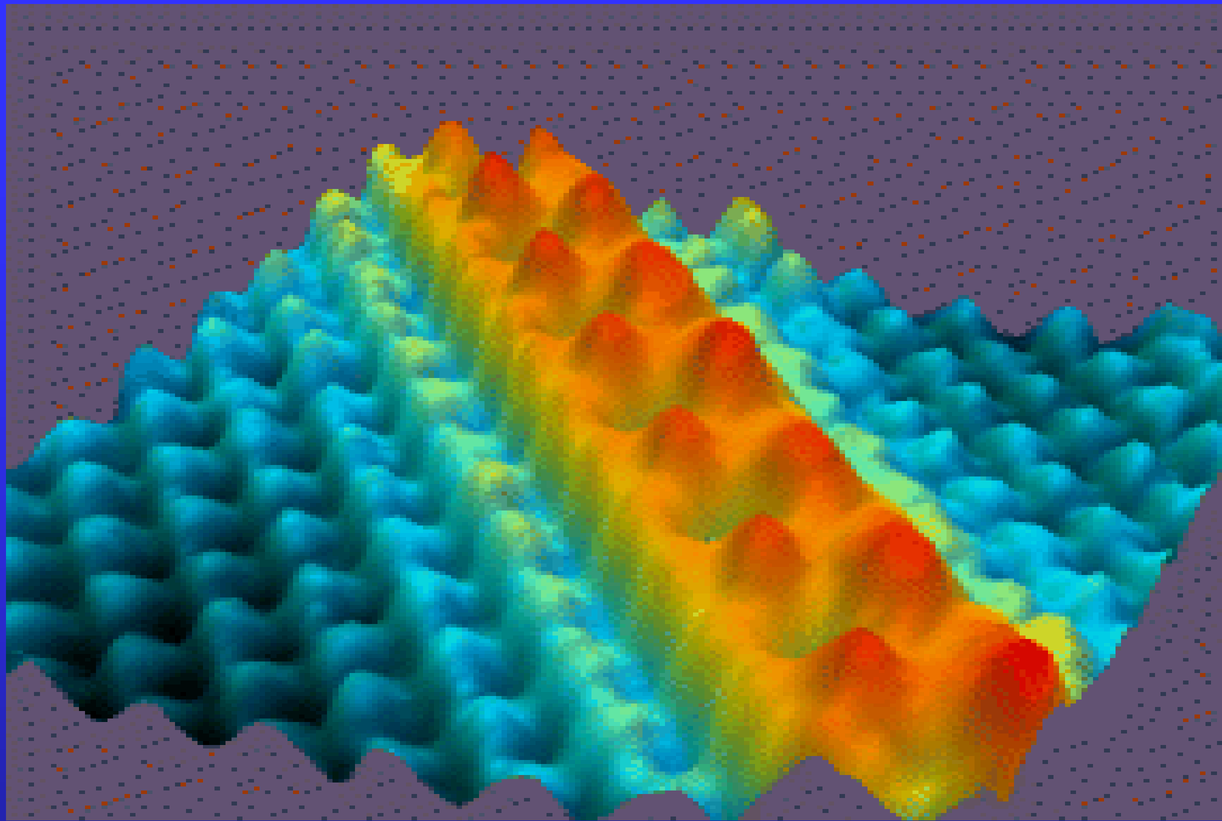
Scanning tunneling microscope (STM)



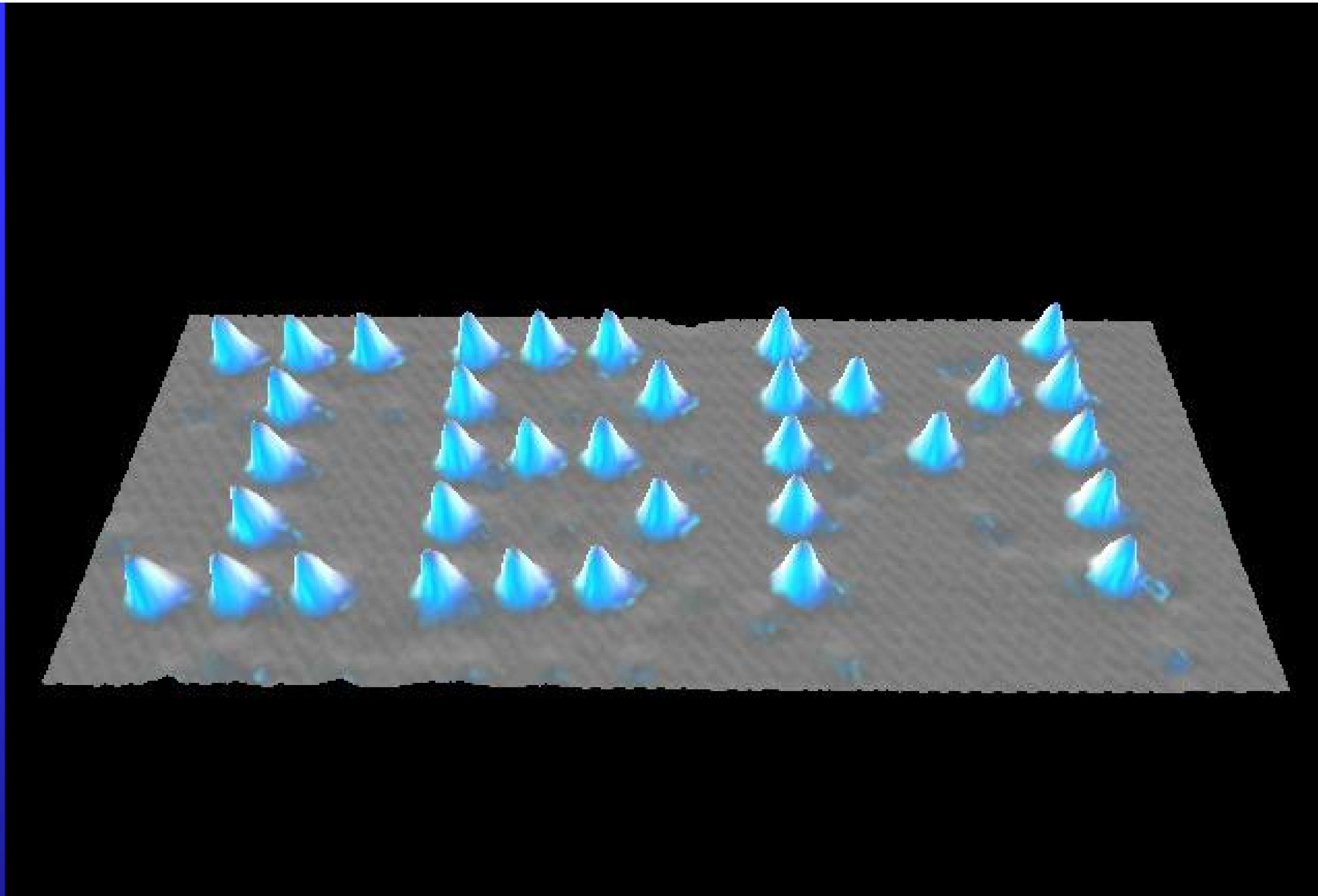
Gerd Binnig and Heinrich Rohrer won the 1986 Nobel Prize due to their STM.



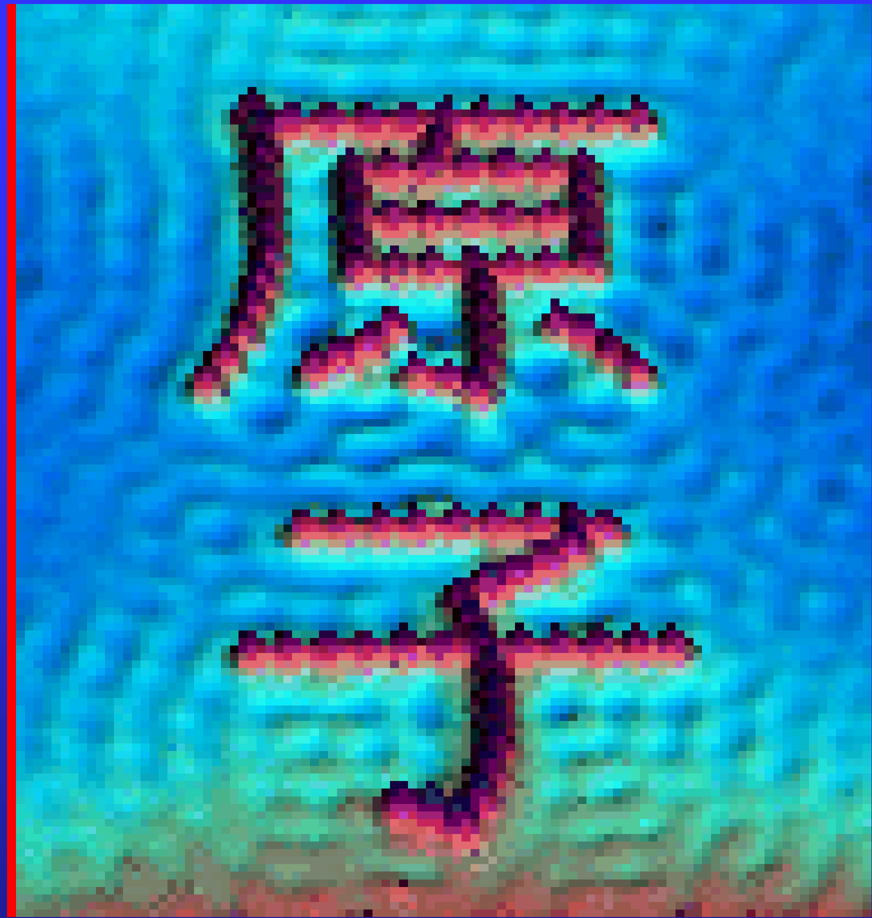
Quantum Corral   Iron on Copper



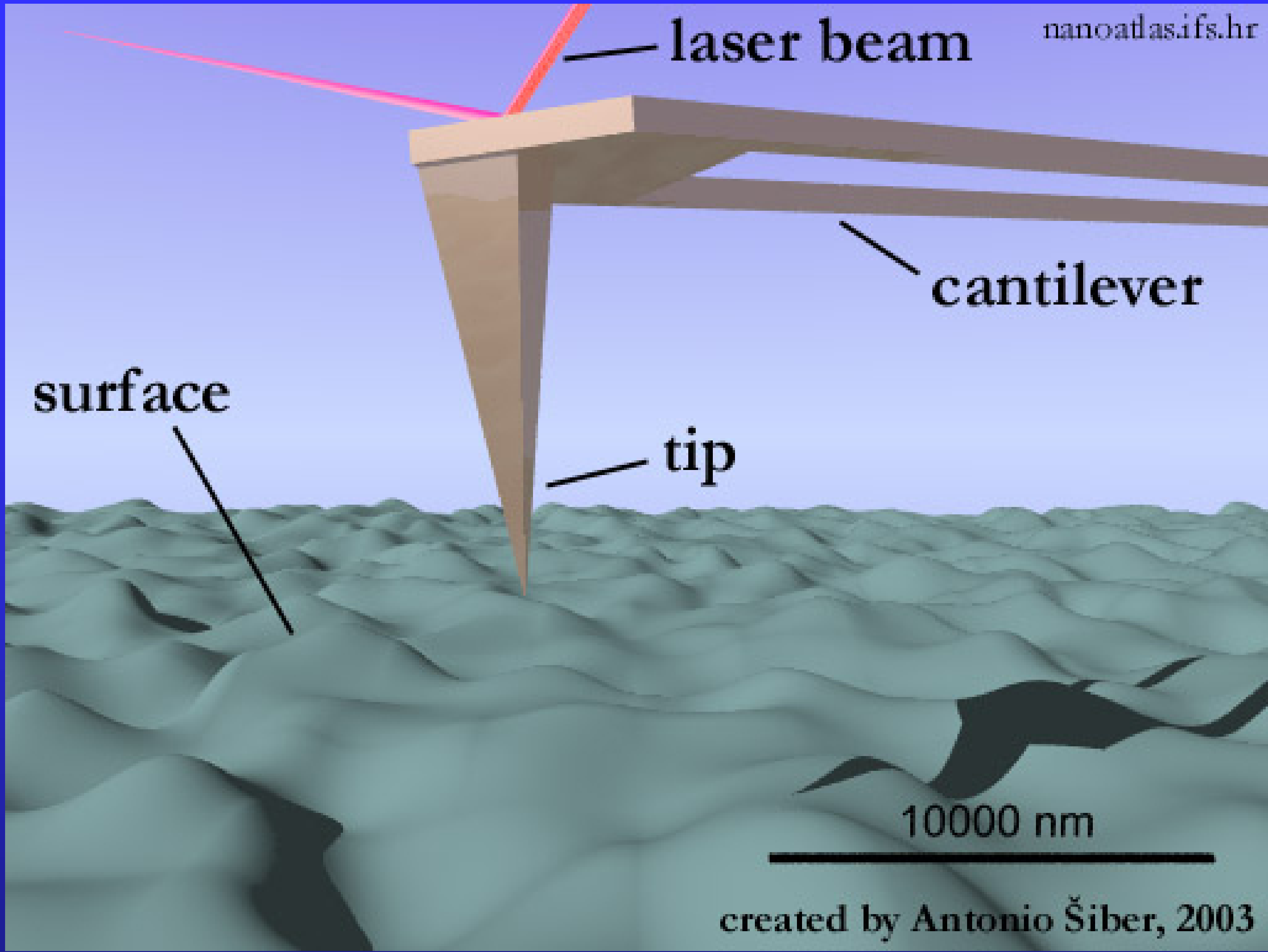
STM image, 7 nm x 7 nm, of a single zig-zag chain of Cs(Cesium) atoms (red) on the GaAs (110) surface (blue). (GaAs Gallium Arsenide)



Xenon on Nickel



Iron on Copper



Atomic force microscope (AFM)

Scanning tunneling microscope (STM)

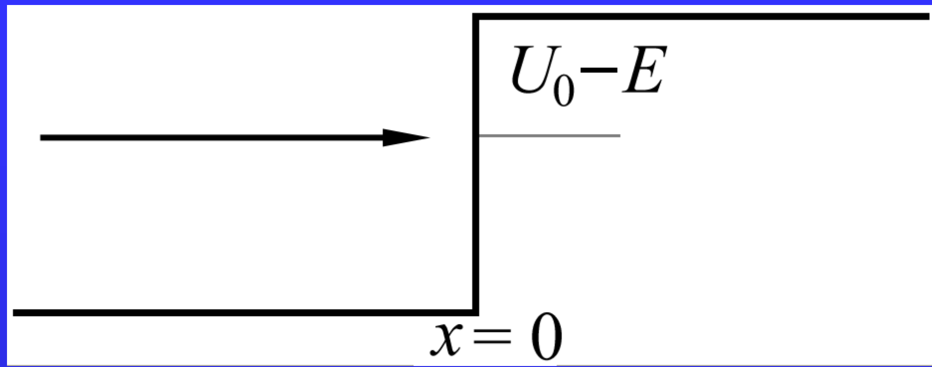
Atomic force microscope (AFM)

Magnetic force microscope (MFM)

Scanning near-field optical microscope (SNOM)

…………

Potential step



$$U(x) = U_0 \theta(x)$$

For $E > U_0$

$$\psi = Ae^{ikx} + Be^{-ikx} \qquad (x < 0)$$

$$\psi = Ce^{ik_1 x} \qquad (x > 0)$$

where

$$k^2 = \frac{2mE}{\hbar^2}, \qquad k_1^2 = \frac{2m(E - U_0)}{\hbar^2}$$

For $E < U_0$,

$$\psi = Ae^{ikx} + Be^{-ikx} \qquad (x < 0)$$

$$\psi = \qquad De^{-\kappa x} \qquad (x > 0)$$

where

$$\kappa^2 = \frac{2m(U_0 - E)}{\hbar^2}$$

Using matching condition at $x = 0$

$$\psi_l\big|_{x=0} = \psi_r\big|_{x=0}$$

$$\left.\frac{d\psi_l}{dx}\right|_{x=0} = \left.\frac{d\psi_r}{dx}\right|_{x=0}$$

a complete solution is possible.

the property of the solution

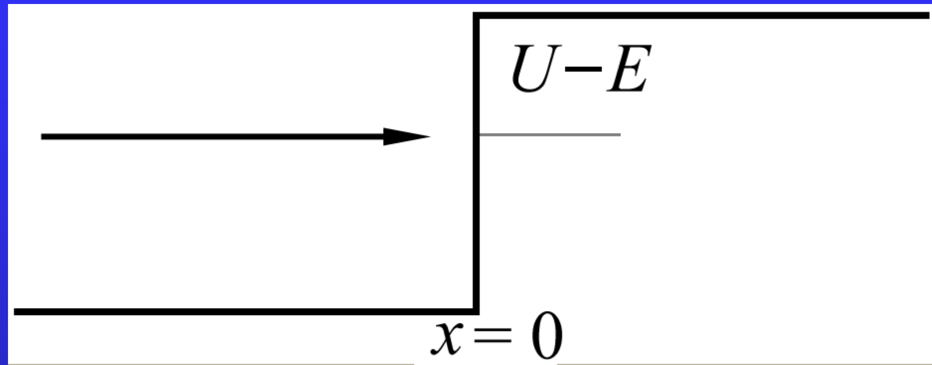
Within some distance ($x > 0$), wave function is still finite ($D \neq 0$), i.e.,

$$|\psi|^2 \sim e^{-2\kappa x} \sim e^{-1}$$

$$\Delta x \sim \frac{1}{2\kappa} = \frac{\hbar}{2\sqrt{2m(U_0 - E)}}$$

According to classical physics, it is impossible to find the particle at $x > 0$.

To overcome potential energy step, the particle must 'borrow' an energy $U_0 - E + T$, and at least $T \sim 0^+$.



The borrowed energy must be returned within time interval $\Delta t = \hbar/(U_0 - E + T)$. So the distance the particle can travel is

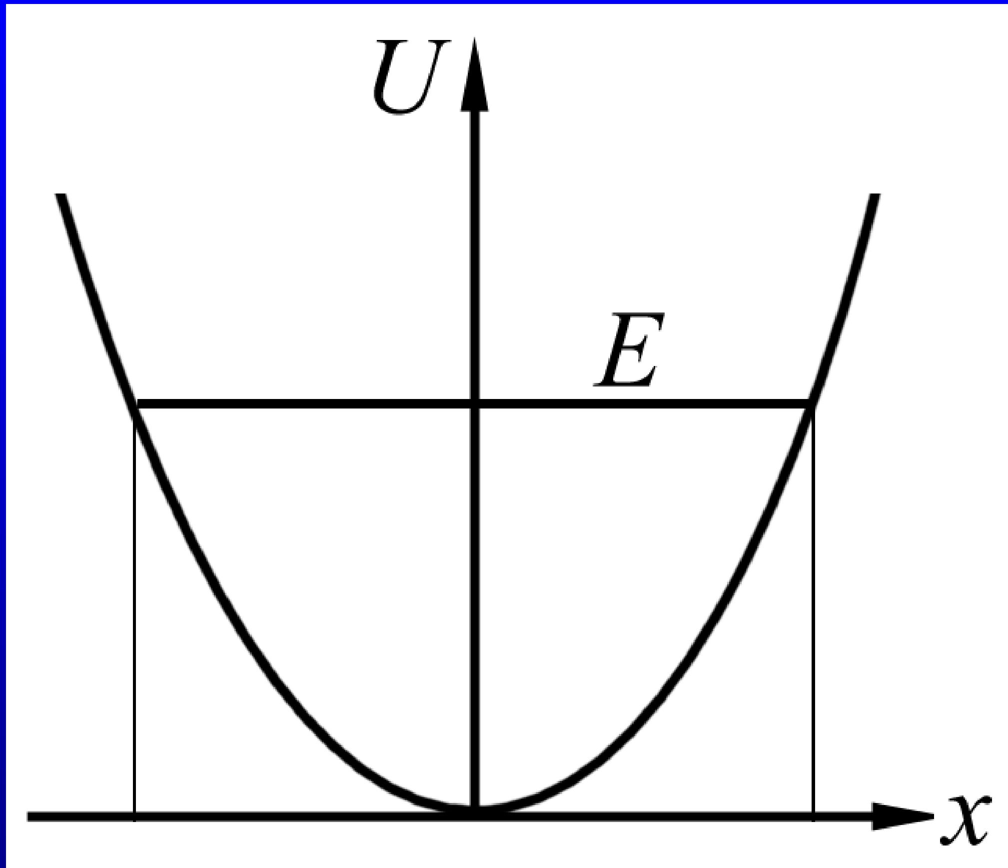
$$\Delta x = \frac{1}{2} v \Delta t = \frac{1}{2} \sqrt{\frac{2T}{m}} \frac{\hbar}{U_0 - E - T}.$$

$$T = 0, \Delta x = 0,$$

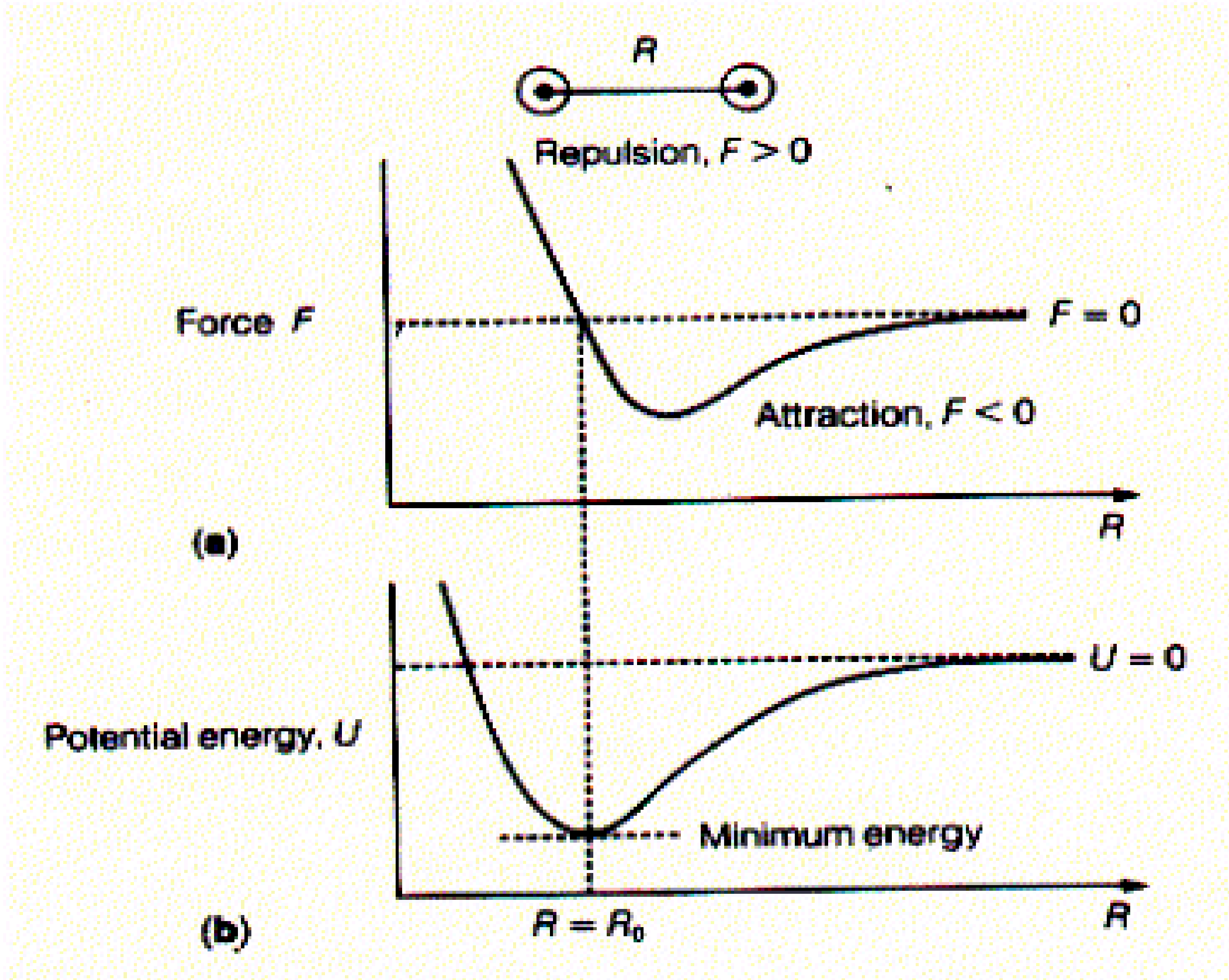
$$T \to \infty, \Delta x_{\max} = \frac{1}{2}\frac{\hbar}{\sqrt{2m(U_0 - E)}}.$$

Assignment: <u>24.5</u>

24.5 Simple harmonic oscillator



$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}kx^2$$



$$U(R) = U(R_0) + U'(R_0)(R - R_0) + \frac{1}{2}U''(R_0)(R - R_0)^2 + \cdots$$

The stationary Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2}+\frac{1}{2}kx^2\psi = E\psi$$

A reasonable requirement is $\psi\ (x \to \pm\infty) \to 0$

(bound state)

<u>trial wave function</u>: $\psi(x) = A\exp(-a\,x^2)$

$$\psi'(x) = -2axAe^{-ax^2} = -2ax\,\psi(x)$$

$$\psi''(x) = -2aA\ (1-2ax^2)\ e^{-ax^2} = \underline{-2a\ (1-2ax^2)\ \psi(x)}.$$

$$\frac{\hbar^2 a}{\underline{m}}(1-2\underline{ax^2})\psi + \frac{1}{2}\underline{kx^2}\psi = \underline{E}\psi.$$

$$\begin{cases} E = \dfrac{\hbar^2 a}{m} \\ -\dfrac{2\hbar^2 a^2}{m} + \dfrac{1}{2}k = 0 \end{cases}$$

$$a = \frac{1}{2\hbar}(km)^{1/2} = \frac{m}{2\hbar}\sqrt{\frac{k}{m}} = \frac{m\omega_0}{2\hbar} \equiv \frac{1}{2}\alpha^2$$

$$\begin{cases} E_0 = \dfrac{1}{2}\hbar\omega_0 \\ \psi_0 = Ae^{-\frac{1}{2}\alpha^2 x^2} \end{cases}$$

ground state

the exact solution

$$E_n = \hbar\omega_0\left(n + \frac{1}{2}\right)$$

$$\psi_n(x) = \left(\frac{\alpha}{2^n n!\sqrt{\pi}}\right)^{1/2} \mathrm{H}_n(\alpha x)\mathrm{e}^{-\frac{1}{2}\alpha^2 x^2}, \qquad \alpha = \left(\frac{m\omega_0}{\hbar}\right)^{1/2}$$

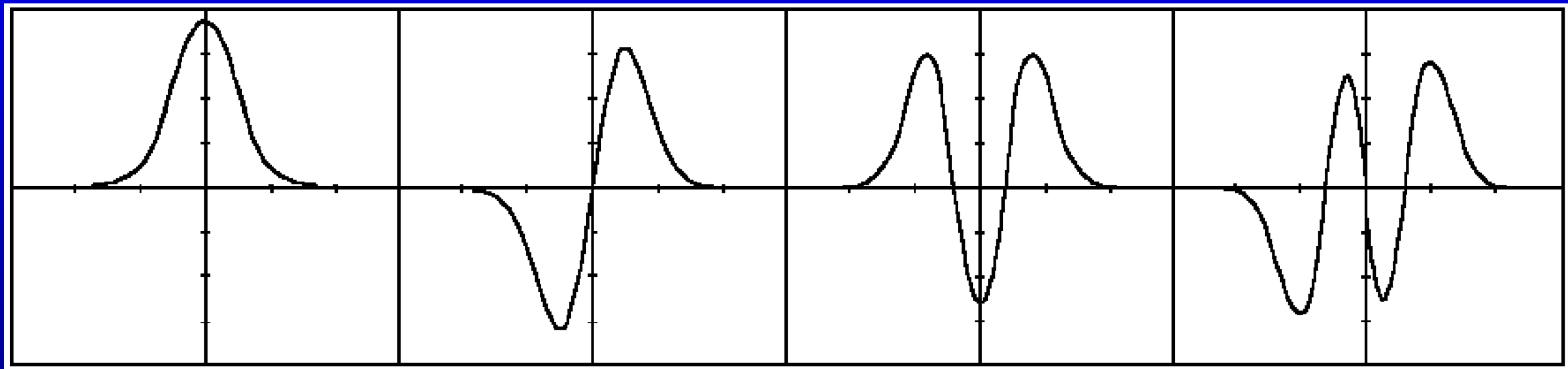
Hermite polynomial

$$\xi = \alpha x$$

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n}{d\xi^n} e^{-\xi^2}$$

$$H_0 = 1, \qquad H_1 = 2\xi$$

$$H_2 = 4\xi^2 - 2, \quad H_3 = 8\xi^3 - 12\xi$$



Peak positions mean max(probability density)

What is the probability density of a classical SHO?

classical SHO

$$P(\xi \to \xi + d\xi) = w(\xi)d\xi = \frac{2dt}{T}; \xi \equiv \alpha x$$

$$w(\xi) = \frac{2}{T\dfrac{d\xi}{dt}}$$

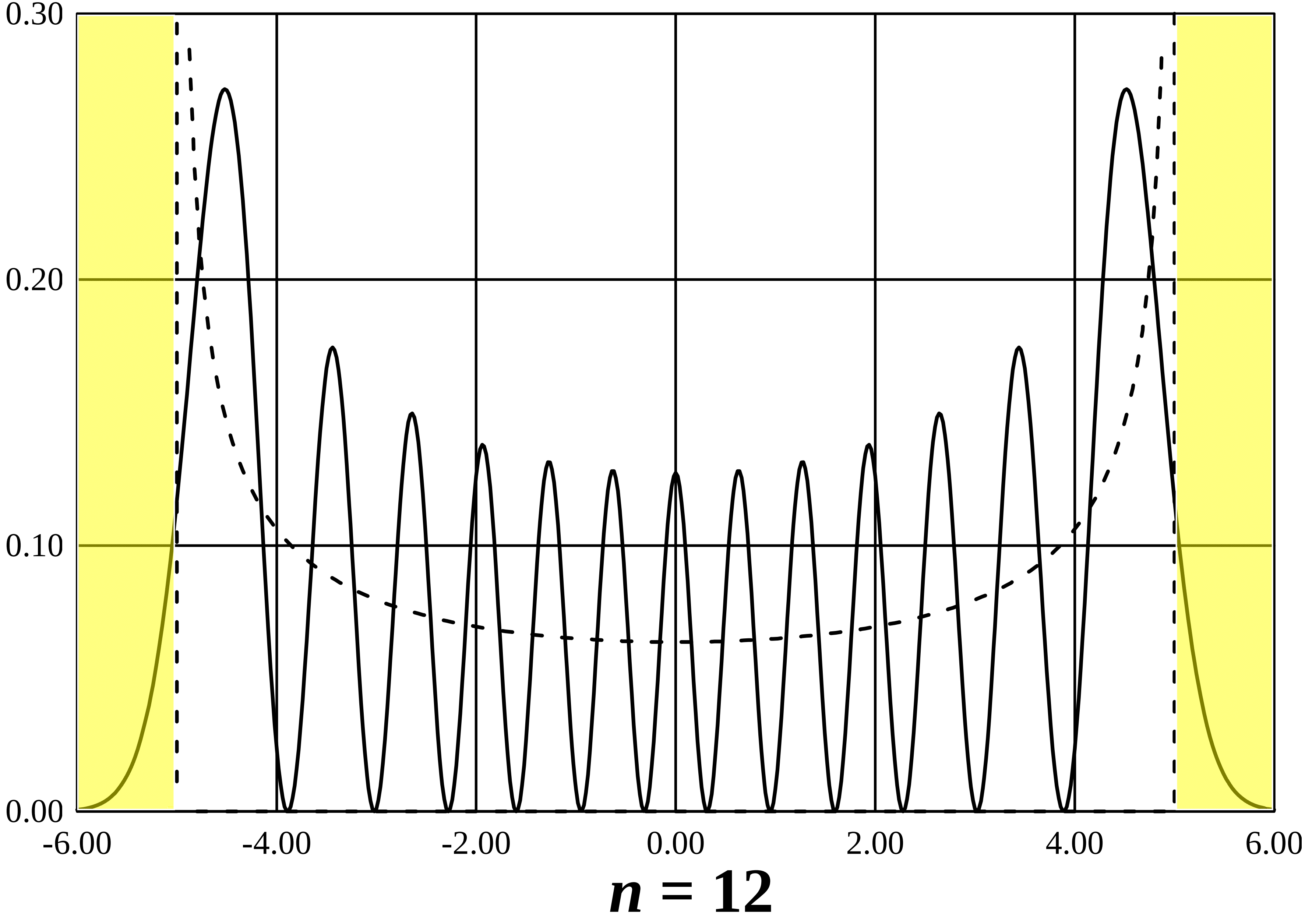
$$\xi = \alpha x = \alpha A \sin(\omega t + \varphi)$$

$$\dot{\xi} = \omega\alpha A \cos(\omega t + \varphi) = \omega\alpha A\left[1 - \sin^2(\omega t + \varphi)\right]^{1/2}$$

$$= \omega\alpha A\left(1 - \frac{\xi^2}{\alpha^2 A^2}\right)^{1/2}$$

The probability is

$$w(\xi) = \frac{1}{\pi \alpha A \left(1 - \frac{\xi^2}{\alpha^2 A^2}\right)^{1/2}}$$


0.30
0.20
0.10
0.00
-6.00
-4.00
-2.00
0.00
2.00
4.00
6.00
n = 12

The ground state energy and wave function are

$$E_0 = \frac{1}{2}\hbar\omega_0$$

zero-point energy

$$\psi_0(x) = \left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} \mathrm{e}^{-\frac{1}{2}\alpha^2 x^2}, \qquad \alpha = \left(\frac{m\omega_0}{\hbar}\right)^{1/2}$$

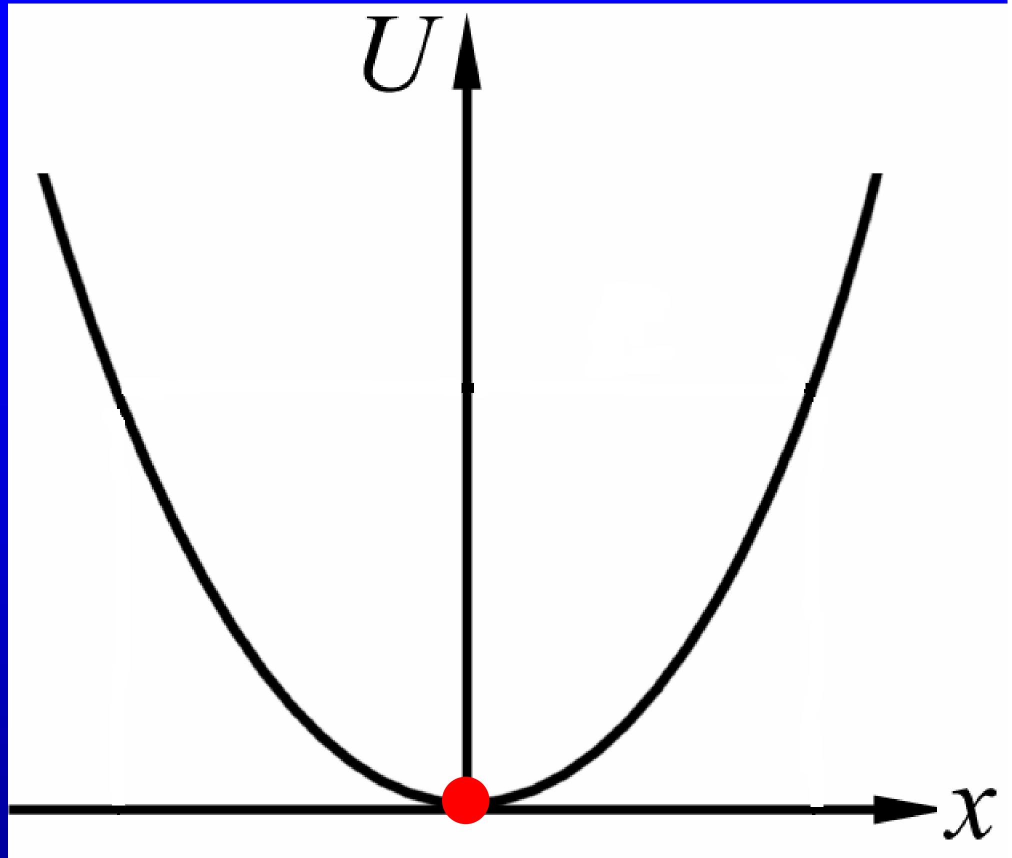
zero-point oscillation

Why zero-point energy?

Why zero-point energy?

For classical 1D harmonic oscillator, if x=p=0, E=0.

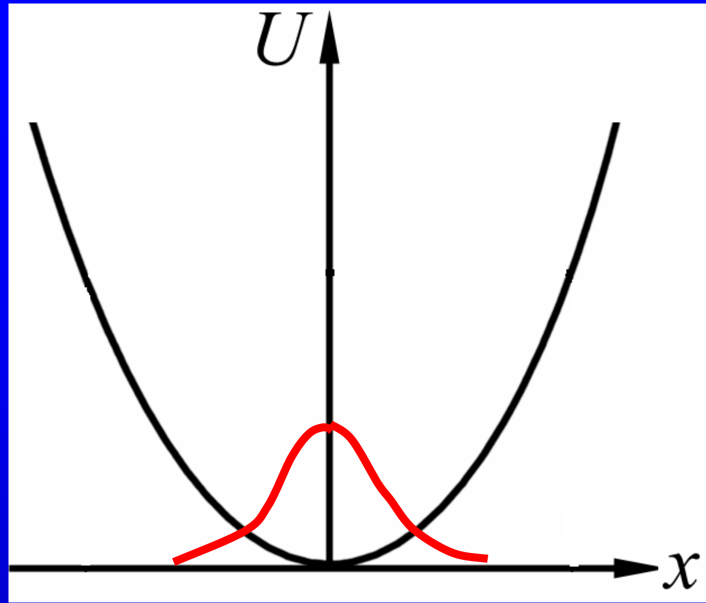
But in the quantum case, due to the uncertainty relation, it is not possible.



In fact,

$$\langle V \rangle = \int \psi_0^*(x)\left(\frac{1}{2}m\omega_0^2 x^2\right)\psi_0(x)dx$$

$$= \frac{1}{4}\hbar\omega_0$$



$$\langle T \rangle = \int \psi_0^*(x)\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\right)\psi_0(x)dx$$

$$= \frac{1}{4}\hbar\omega_0$$

$E_0 = \frac{1}{2}\hbar\omega_0$ is the lowest energy possible.

24.6 The formal theory of quantum mechanics

The Schrodinger equation

$$3D: \quad i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + U(\boldsymbol{r})\psi.$$

$$1D: \quad i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi + U(x)\psi.$$

$$3D: \quad \hat{\boldsymbol{p}} = \frac{\hbar}{i}\nabla, \quad 1D: \ \hat{p} = \frac{\hbar}{i}\frac{d}{dx}.$$

Stationary Schrodinger equation

$$\Psi(\boldsymbol{r},t) = \psi(\boldsymbol{r}) f(t).$$

$$\hat{H}\psi = E\psi. \qquad \psi = \psi(\boldsymbol{r})$$

$$3D: \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + U(\boldsymbol{r})$$

$$1D: \hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)$$

Several principles (postulates) of quantum mechanics:

1. To every physical quantity $Q(\boldsymbol{r}, \boldsymbol{p})$, , there corresponds a quantum operator obtained by replacing $\boldsymbol{p}$ with $\frac{\hbar}{i}\nabla$, , i. e. $Q(\boldsymbol{r}, \frac{\hbar}{i}\nabla)$.

quantities	classical definition	quantum operator
position	$\boldsymbol{r}$	$\boldsymbol{r}$
momentum	$\boldsymbol{p}$	$\frac{\hbar}{i}\nabla$
angular momentum	$\boldsymbol{r}\times\boldsymbol{p}$	$\frac{\hbar}{i}\boldsymbol{r}\times\nabla$
kinetic energy	$\frac{\boldsymbol{p}^2}{2m}$	$-\frac{\hbar^2}{2m}\nabla^2$
total energy	$\frac{\boldsymbol{p}^2}{2m}+U(\boldsymbol{r})$	$-\frac{\hbar^2}{2m}\nabla^2+U(\boldsymbol{r})$

$$\hat{Q} = Q(r, \frac{\hbar}{i}\nabla)$$

eigenvalue

$$\hat{Q}\psi = q\psi$$

eigenequation

eigenfunction

possible eigenvalues: $q_1, q_2, q_3, \cdots$

corresponding eigenfunctions $\psi_1, \psi_2, \psi_3, \cdots$

2. The only possible values that can be obtained when a physical quantity Q is measured are the eigenvalues of the quantum operator $\hat{Q}$.

$$\hat{Q}\psi_n = q_n\psi_n$$

$$\int \psi_n^* \psi_m d\tau = \delta_{nm.}$$

Each of $\{\psi_n\}$ is normalized and orthogonal to each of the others.

an arbitrary quantum states

$$\phi = \sum_n c_n\psi_n = c_1\psi_1 + c_2\psi_2 + \cdots$$

$$c_n = \int \psi_n^* \phi d\tau.$$

3. When a state of the system corresponds to the wave function $\phi(\boldsymbol{r})$, the possibility of obtaining value q_n as a result of a measurement of the physical quantity Q is $|c_n|^2$ where c_n is given by

$$\phi = \sum_n c_n \psi_n$$

and ψ_n is the eigenfunction of the operator $\hat{Q}$ corresponding to the eigenvalue q_n .

4. The time evolution of the physical system is described by the equation

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H}\Psi$$

where $\hat{H}$ is the Hamiltonian operator of the system.

Assignment: 24.6, 24.7